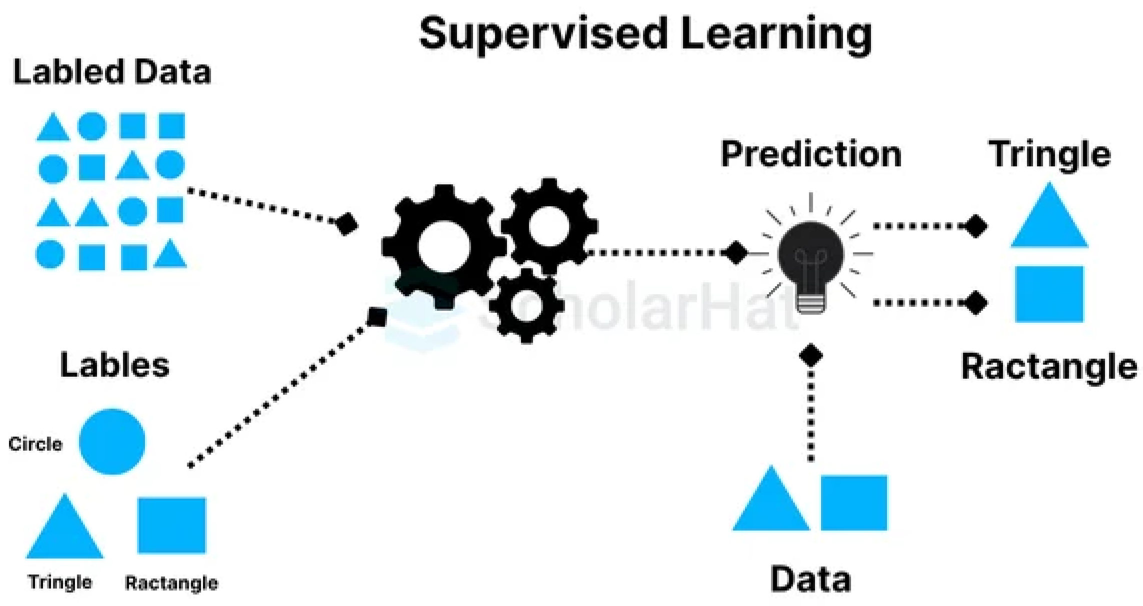
Supervised Learning in AIoT



What is Supervised Learning?

📚 Learning from labelled data
✓ Model learns input → output relationships

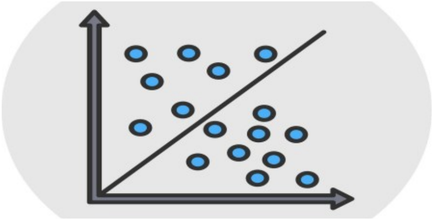
How It Works

- Provide historical labelled examples
- Train model to recognize patterns
- Predict outcomes for new data

AIoT Use Cases

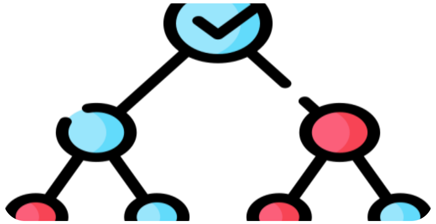
🏭 Predict equipment failure
🌡 Forecast temperature & energy demand
🚦 Classify device operating states
🚨 Detect safety violations

common Supervised Learning Algorithms in AIoT



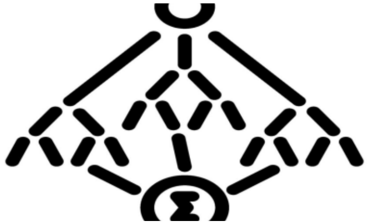
Logistic Regression

- Used for binary classification
- Predicts probability of an event
- **Example:** Predict whether a machine will fail (Yes/No)



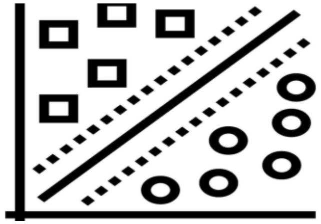
Decision Trees

- Rule-based decision structure
- Easy to interpret & visualize
- **Example:** Classify device state: Normal → Warning → Fault



Random Forest

- Ensemble of multiple decision trees
- Improvs accuracy & reduces overfitting
- **Example:** Predict equipment failure using multiple sensor inputs

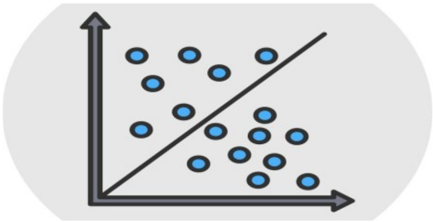


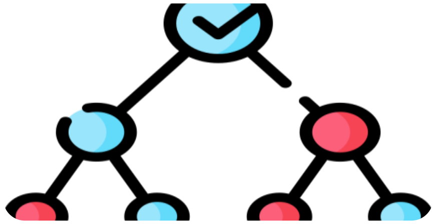
Support Vector Machines (SVM)

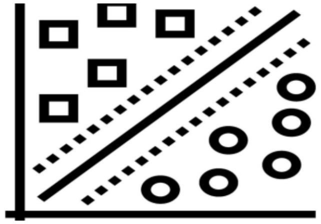
- Finds optimal boundary between classes
- Effective for complex classification problems
- **Example:** Detect anomaly vs normal sensor behaviour

These algorithms enable reliable prediction and classification in AIoT systems

When to Use Which Algorithm in AIoT







Logistic Regression

- Binary outcomes & probability prediction
- Simple, fast, and interpretable
- **Use when:**
  ✓ Predict failure vs normal
  ✓ Estimate risk probability
  ✓ Limited compute resources

Decision Trees

- Rule-based decisions
- Easy to understand & explain
- **Use when:**
  ✓ Need explainable decisions
  ✓ Device state classification
  ✓ Operational decision logic

Random Forest

•Higher accuracy & robustness

•Handles multiple sensor inputs

•Use when:
✓ Predictive maintenance
✓ Complex sensor relationships
✓ Need better accuracy than singie tree

Support Vector Machines (SVM)

•Clear separation between classes

•Effective with complex patterns

•Use when:
✓ Anomaly detection
✓ Fault vs normal classification
✓ Moderate-sized datasets

Choose based on complexity, interpretability, accuracy needs, and device constraints

Real-Life AIoT Use Case: Smart Building Energy & Equipment Monitoring

🎯 **Scenario :** A smart commercial building uses IoT sensors to monitor HVAC systems, temperature, energy consumption, and equipment health to improve efficiency and prevent failures.

1️⃣ Logistic Regression → Failure Risk Prediction

- **Problem :** Will an air-conditioning compressor fail in the next 24 hours?
- **Why Logistic Regression?**
  ✓ Predicts probability of failure
  ✓ Lightweight & efficient for edge devices
- **Outcome:** Maintenance team receives early risk alerts.

2️⃣ Decision Tree → Equipment State Classification

- **Problem :** What is the current operating state of HVAC equipment?
- **Possible States :** Normal → Overloaded → Fault
- **Why Decision Tree?**
  ✓ Easy to interpret rule-based decisions
  ✓ Helps facility teams understand cause
- **Outcome :** Operators see clear decision logic and act quickly.

Real-Life AIoT Use Case: Smart Building Energy & Equipment Monitoring

3 Random Forest → Energy Optimization

- **Problem :** How can energy consumption be optimized without affecting comfort?
- **Why Random Forest?**
  ✓ Handles multiple variables (temperature, occupancy, humidity, time)
  ✓ Provides accurate predictions
- **Outcome:** System automatically adjusts cooling schedules → saves energy costs.

4 Support Vector Machine → Anomaly Detection

- **Problem :** Is equipment behaviour abnormal compared to normal patterns?
- **Why SVM?**
  ✓ Excellent at separating normal vs abnormal patterns
  ✓ Works well with sensor data patterns
- **Outcome :** Early detection of abnormal vibration or pressure prevents breakdown.

Business Impact

- Reduced unplanned downtime
- Improved production efficiency
- Lower maintenance costs
- Higher product quality
- Increased operational safety

Real-Life AIoT Use Case: Smart Factory Predictive Maintenance & Quality Control

🎯 **Scenario :** A manufacturing plant uses IoT sensors on machines to monitor vibration, temperature, pressure, and production quality to prevent downtime and ensure product consistency.

1️⃣ Logistic Regression → Failure Risk Alert

- **Problem :** Is a machine likely to fail in the next few hours?
- **Why Logistic Regression?**
  ✓ Predicts probability of failure
  ✓ Fast & lightweight for edge deployment
- **Outcome :** Maintenance teams receive early warning alerts before breakdown occurs.

2️⃣ Decision Tree → Machine Operating State

- **Problem :** What is the current machine condition?
- **Possible States :** Normal → Overheating → Critical Fault
- **Why Decision Tree?**
  ✓ Easy to interpret rule-based classification
  ✓ Helps technicians understand root cause
- **Outcome :** Operators can quickly diagnose and respond.

Real-Life AIoT Use Case: Smart Building Energy & Equipment Monitoring

3 Random Forest → Predictive Maintenance

- **Problem :** When should the machine be serviced?
- **Data Used :** Vibration + temperature + load + runtime hours
- **Why Random Forest?**
  ✓ Combines multiple sensor inputs
  ✓ Provides high accuracy & reliability
- **Outcome :** Maintenance is scheduled before failures, reducing downtime.

4 Support Vector Machine → Quality Defect Detection

- **Problem :** Are produced items defective based on sensor & vision data?
- **Why SVM?**
  ✓ Effective for separating acceptable vs defective patterns
  ✓ Works well with pattern recognition
- **Outcome :** Defective products are detected early, reducing waste.

Business Impact

Reduced unplanned downtime	Improved production efficiency	Lower maintenance costs	Higher product quality	Increased operational safety

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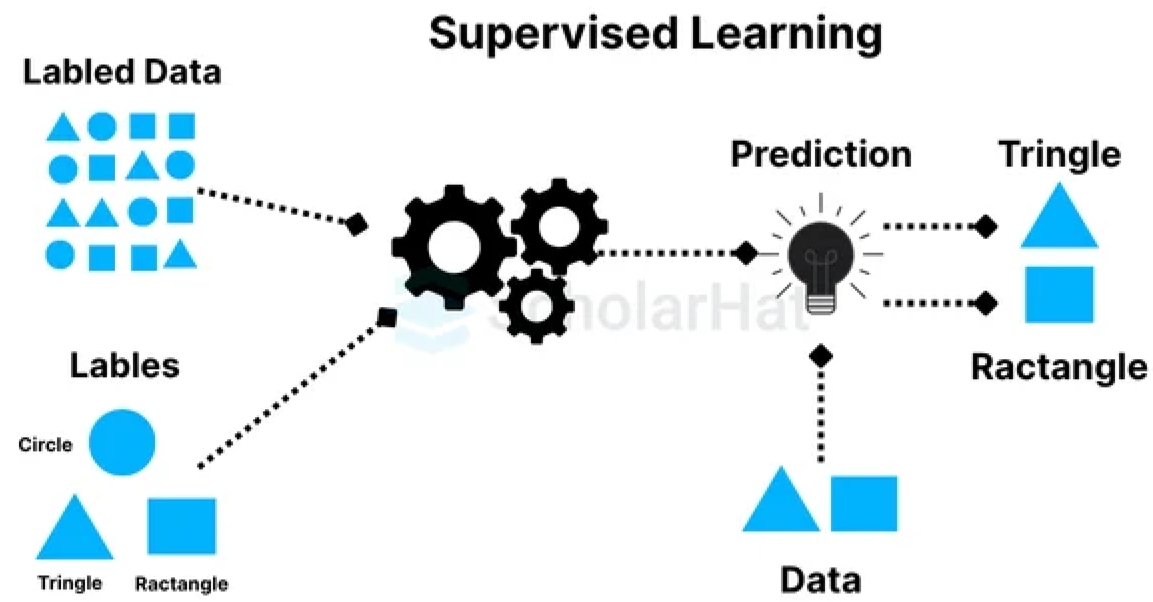

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